

Buddy Collins

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Education

University of Colorado | Boulder, CO

Master of Science in Network Engineering

Expected May 2028

Graduate Certificate in Project Management

GPA: 3.7/4.0

- Focus: Cybersecurity, LAN/WAN Architecture, Network Reliability, Cloud Networking.

University of Colorado | Boulder, CO

Bachelor of Arts in Geography, Geospatial Information Systems (GIS) emphasis

May 2020

- Graduate coursework in Geography and Geospatial Information systems
- Focus: GIS, Spatial Data Analytics, Automation, Geodatabase Management.

Professional Experience

National Center for Atmospheric Research (NCAR) | Boulder, CO

Network Engineer

July 2020 – Present

- Architect, deploy, and maintain research-grade wide-area-network (WAN) and local area network (LAN) infrastructures, core routing platforms, and out of band (OOB) management networks to ensure secure management paths and resilient network administration, maintaining 99% uptime for NCAR Facilities and 31 Front Range GigaPop (FRGP) Partners.
- Direct field engineering teams in fiber deployment, OTDR signal verification and emergency restoration activities to resolve physical and transport layer network faults and infrastructure risks.
- Perform high precision network testing, diagnostics, and fault isolation across 724+ miles of fiber using OTDRs, OSAs, optical power meters, fiber scopes and other industry-standard network troubleshooting tools to isolate system faults, detect signal anomalies and verify physical layer network integrity.
- Perform system patching and device configuration management across multi-vendor network infrastructure to minimize operational risk and prevent unauthorized access.
- Document/author network architectures, fiber routes, and disaster recovery procedures to ensure operational compliance and rapid incident response.
- Lead GIS initiatives 10+ interactive web applications, cartographic representations, and enterprise geodatabases to audit and monitor physical network assets.
- Mentored and instructed 8 student interns on GIS software, network diagnostic tools and technical documentation processes.

Student Assistant

July 2019 – July 2020

- Supported senior engineers during critical outages, executed emergency remote hands support, physical fiber deployments and inventory tracking.
- Engineered custom ArcGIS online web mapping applications to model physical infrastructure.

Affiliations & Leadership

SCinet (Super Computing Network Team)

WAN Team Member (SC23)

Fall 2023

- Collaborated with international engineers to design, deploy and maintain the world's fastest temporary research network supporting Super Computing conference infrastructure.

FRGP (Front Range Giga-Pop)

Wide Area Network Engineer

2021 – Present

- Design, deploy, document and maintain non-profit research area infrastructure in direct collaboration with senior network engineering leads.

WINS (Women in Networking at SC)

Digital Engagement lead

2023 – 2025

- Managed web presence and outreach programs to expand visibility for diversity and inclusion initiatives within the high-performance computing and SCinet communities

Technical Skills and Competencies

- Cyber & Network infrastructure: Cisco, Juniper, Adtran, MikroTik, WAN/LAN Architecture, Fiber optics, OOB Management, routing Protocols, OTDR, OSA, Optical Power Meters, Visual Fault Locators.
- Programming & Systems: Python, C++, JavaScript, HTML, CSS, R, GitHub, Linux/Bash automation.
- Geospatial & data Analytics, ArcGIS Suite (Pro, Online, ArcPy), QGIS, GDAL, OGR, Geopandas, Pandas, GeoJSON, KMZ, Leaflet, Fiona.
- Software & Documentation: Enterprise Systems Documentation, Microsoft Office Suite, Google workspace.